

x2 Modem Technical Specification

Negotiation downstream mapping and symmetric operation

Draft 0.7 • Independent firmware reconstruction • 4 October 2026

1 Scope and operating model

This document explains x2 negotiation and data mapping for asymmetric and symmetric operation. It covers the Courier 3453C analogue client, original I-modem server and later Quad mapper evidence. Clauses 12 to 14 describe symmetric operation and the successful digital pair. Clauses 15 to 17 cover the full rate ladder, lower-rate mapping, scrambling and the limits of symmetric attribution.

The client retains executable x2 setup and DSP code. The server capability exchange and sample transforms have been reconstructed. The later Quad reconstruction connects received parameters to six mapping banks and provides an execution-checked source-byte-to-codeword core with amplitude selection and sign shaping. Stock controller copies validate overlay placement. Complete peer negotiation and analogue client decoding remain open.

In an asymmetric connection, the digital server supplies downstream sample codewords. The analogue client receives the resulting waveform and must recover the information represented by the chosen levels. The upstream direction uses the related V.34 machinery. In symmetric operation both endpoints are digitally attached and send and receive at the same x2 rate. A paired emulator test establishes 64000/64000 carrier and bidirectional marker delivery; the complete symmetric negotiation and data transition have not yet been reconstructed.

A constellation is the set of signal levels available for carrying data. A mapper selects members from input data; the receiver performs the inverse. A level table alone does not establish framing, shaping, synchronization or payload decoding.

2 Status of the specification

The numbered clauses and field tables follow the style of an ITU technical specification. This is an independent engineering draft, not an ITU-T Recommendation. Statements describe the recovered firmware profile; they are not universal requirements for all x2 equipment.

Table 1 — Evidence used in this draft

Designation	Meaning
Verified	A semantic reconstruction matches original DSP instructions, or a stated behavior passes a recorded emulator call test. The qualification identifies which evidence applies.
Observed	Firmware instructions, constants or message construction establish a local operation without validating its complete protocol procedure.
Provisional	Component behavior is reproducible, but a placement assumption or protocol interpretation remains unverified.
Open	The available evidence does not yet establish the procedure.

Annex A collects firmware references, memory layouts and reproduction commands. The main clauses describe connection behavior and signal parameters. Timing and interoperability requirements are stated only where the evidence supports them.

3 Negotiation sequence

Table 2 organizes the recovered operations into a connection sequence. The individual operations are supported to different degrees; their complete ordering and transition timing have not been demonstrated in one end-to-end call.

Table 2 — Recovered negotiation stages

Stage	Purpose and observed behavior	Status
Local selection	Check whether x2 is enabled and permitted by the selected mode, speed settings and line conditions. Prepare its capability announcement.	Observed
Capability exchange	Use the INFO0 capability structure to announce permitted asymmetry and related attributes. The server route into the outgoing capability buffer is verified.	Verified in server
Directional announcement	Transmit a short proprietary body naming a carrier choice and two modulation-rate indices. One index can identify the PCM direction.	Observed
Channel qualification	Evaluate peer parameters and a signed channel margin. Related rate machinery limits the requested rate against available capability.	Partial
Training and probing	Generate a repeating startup source and collect received training words and a position bitmap.	Source verified
Constellation preparation	Decode received choices, select candidate level sets and expand six position banks.	Components verified
Data operation	Map source bytes to six codewords with differential signs and disparity shaping. An ideal symbol inverse recovers the bytes; analogue client decoding and call alignment remain open.	Core verified

3 1 Local selection and fallback

The client calls a distinct x2 eligibility procedure during connection setup. It excludes disabled x2 operation, unsuitable modem modes and speed or line conditions that fail the firmware gates. The neighboring V.90 selection procedure is separate. Retaining an x2 setup call establishes an active selection mechanism, but does not prove successful peer negotiation.

The setup prepares a capability pattern describing directional asymmetry. Some settings can suppress that asymmetry. Equal modulation indices in a later announcement establish equal directional parameters, but do not by themselves identify x2 symmetric operation: the equal pair shown in Table 5 uses ordinary V.34 index 4. Symmetric x2 has separate role capability controls, described in clause 12. The precise transition from rejected x2 operation to an established fallback connection is still open.

3 2 What remains to connect

The client stores the x2 capability supplied by its supervisor, but the consumer that connects that particular store to the transmitted capability buffer has not been traced. Its tested initialization fragment reads a different capability word. This missing connection is specific to the 3453C profile and must be resolved before claiming complete client negotiation.

4 Capability and modulation announcements

4 1 INFO0 capability fields

The interpreted client capability announcement uses fields already present in the V.34 INFO0 structure. The supervisor sets the permitted asymmetry, clears the CME indication and clears the clock-source field. Table 3 states the interpreted wire field correspondence. The server capability word has a verified route into its INFO0 work buffer; actual frame emission and peer acceptance remain separate checks.

Table 3 — Interpreted capability fields

INFO0 bits	Parameter	Client construction
21 to 23	Maximum directional symbol-rate difference	An integer from 0 to 3, selected by the two relevant S58 settings
24	CME indication	Cleared
26 to 27	Transmit clock source	Cleared

With both relevant S58 options clear, the asymmetry integer is 2. Enabling the first option alone makes it 3; enabling the second alone makes it 0; enabling both makes it 1. These are numerical values, not printed bit strings that might be confused with transmission order.

4 2 Directional modulation parameters

The proprietary body has seven bits. The lowest bit selects the carrier; the next three bits hold the first modulation index and the highest three hold the second. The roles of the two indices interchange with the modem role flag. Index 6 is interpreted as the PCM-direction indication; ordinary V.34 modulation indices occupy the range 0 to 5.

Table 4 — Directional announcement body

Body bits	Width	Meaning
0	1	Carrier choice: 0 low, 1 high
1 to 3	3	First modulation index
4 to 6	3	Second modulation index

Table 5 — Observed announcement values

Body value	Carrier	First index	Second index
4D hexadecimal	High	6	4
69 hexadecimal	High	4	6
48 hexadecimal	Low	4	4
49 hexadecimal	High	4	4

The first two values announce the same asymmetric pair with its directions exchanged. The latter two name equal ordinary V.34 indices and differ only in carrier choice; they are not proof of a PCM connection in both directions. These values come from the inspected Courier announcement builders; they have not been established as a complete set of every possible x2 peer message.

5 Framing recognition and rate agreement

5 1 Transmitted announcement structure

The recovered x2-only announcement script contains two lead-in ones, twelve bits of fill and synchronization, the seven-bit body, sixteen check bits and four closing ones. The framed portion is 39 bits; including the lead-in gives 41 emitted script bits. A lead-in is therefore not counted as part of the 39-bit frame.

Table 6 — Announcement script in emission order

Element	Bits	Interpretation
Lead-in	2	Two ones before the framed portion
Fill and synchronization	12	Shared INFO framing sequence
Information body	7	Carrier and directional modulation indices
Check field	16	INFO CRC family
Closing fill	4	Four ones

The local field encoder models the body least-significant bit first and the V.34 INFO CRC initialized to all ones, using the reflected form of the polynomial with terms of order 16, 12, 5 and 0. This is a working serialization model. A captured x2 frame or an executed complete serializer is still needed to validate every bit convention.

5 2 Receiver recognition remains open

A seven-bit comparison in the shared receiver was previously interpreted as recognition of the proprietary announcement. That interpretation is superseded: the same comparison exists in pre-x2 V.34 firmware and matches a window of the INFO fill and synchronization preamble. It does not identify an x2 body value.

No receiver for the proprietary seven-bit body has been conclusively identified. The framing script proves construction of an announcement; it does not prove how a server accepts it. The document therefore does not assign an acknowledgement, timeout or retry procedure to this message.

5 3 Channel qualification and rate limits

The 3453C qualification fragment tests a peer attribute and computes a signed margin from peer parameters and a channel measurement. A positive margin enables a subsequent state choice. The physical unit of that margin and the peer identity implied by the test remain open.

Related later firmware compares a host-requested maximum with the existing maximum and retains the smaller value. Its two directional rate fields exchange roles with the same role flag used by the directional announcement. It also intersects enabled rate capabilities with a symbol-rate-dependent mask and the extended-constellation capability. This is evidence of inherited V.34 rate agreement; its complete x2-specific binding remains open.

Repository rate analysis supports a six-symbol downstream organization with nominal rate increments of $8,000 \div 6$ bit/s. The new six-position mapper evidence is consistent with that organization. Consistency does not establish its alignment signal, negotiated bit allocation or exact rate-selection procedure.

6 Startup and receiver qualification

6 1 Repeating startup source

The original server has a four-state startup source. Starting from its initialized state, it repeats the 2,010-call cycle in Table 7. Execution comparison over 4,096 consecutive source calls covers two complete cycles and part of a third.

Table 7 — Verified server source cycle

Part	Source words	Count
Training prefix	Repeated hexadecimal 7E	1,747
Separator	Zero	7
Probe pairs	Ascending first word and descending second word: (0,255), (1,254), continuing to (127,128)	128 pairs
Restart	Return to the training prefix	Cycle repeats

The pair generator advances the ascending value after selecting its return word, then emits the descending value before decrementing it. On the last pair, it restores the prefix count and returns to the first state. The reconstructed delayed-return updates match the original instructions.

These are source words. The selected output path may mask their width, scramble them and reverse an octet before transmission. A source-call count does not establish elapsed time. The sample cadence and callback selected during each startup stage must be traced before specifying an on-wire duration.

6 2 Received training and position bitmap

The inspected receive chain begins by qualifying repeated training values. It advances a position counter over six positions and accumulates a bitmap. Later code classifies received low-valued patterns, records words in a small report buffer and changes receive substates. The initial detector seeks seven qualifying training values before advancing.

The bitmap and recorded reports are plausible inputs to channel qualification and constellation selection, but their full relationship to the outgoing probe pairs is not established. The receiver has not yet been execution compared as a complete sequence. Its report format, rejection conditions and transition deadlines remain open.

6 3 Transition to operating transforms

The server sample path can select seven-bit or eight-bit source words. Transmission applies the selected mask, optional scrambling, optional octet reversal and an output write. Reception records a sample, performs the inverse reversal, applies the mask and optionally descrambles it before delivery.

The tested sample transforms match the original routines with payload source and sink dispatch disabled. They establish transformation behavior, but not supervisor queue delivery or a complete data connection. A seven-bit digital bearer word must not be interpreted as the downstream analogue x2 rate without the mapper and frame allocation.

7 Negotiated downstream parameters

7 1 Transfer of the received record

In the inspected PCM receive branch, four buffered words become the working negotiation record. The first two words supply directional parameters; the last two supply the position choices and an additional parameter. This transfer and the subsequent unpacking match the original instructions. The branch condition is part of the verified entry context; the check does not include frame reception, CRC acceptance or acknowledgement.

Table 8 — Recovered working parameters

Parameter	Recovered representation	Qualification
Original mode	Four bits from the first working word	Used to select a scale table
Effective mode	Original mode, or zero when a local override is set	Used by a second parameter table
Independent sign count	Three bits above the position record	MD from 0 to 6; values outside this range are unverified
Position record	Twenty-four bits organized as six four-bit cells	Three bits of each cell select a control value

The distinction between original and effective mode is observable: a local override changes the stored mode to zero, while one scale-table lookup retains the original mode. The draft therefore keeps both parameters distinct. Their exact rate names and the physical units of the scale values have not been established.

7 2 Conversion of position choices

The firmware consumes the six cells from the most significant end of its working record. It ignores the upper bit of each cell and converts the lower three bits according to Table 9. Each result occupies two bits in the position-control word. Bank expansion then consumes that word from its least significant end, so its first bank corresponds to the last cell processed during unpacking. This describes working-record order; wire order remains to be established.

Table 9 — Cell value to position control

Cell value	0	1	2	3	4	5	6	7
Control	0	1	2	1	0	3	0	3

The firmware also counts the nonzero controls. If all six controls are zero, it substitutes a control word with the lowest position set to value 2, while retaining an active-position count of zero. This is a distinct default state. It must not be described as six ordinary zero choices or silently recounted as one active position.

7 3 Extent of the verified negotiation path

Received-record transfer, unpacking and bank expansion match the original instructions for all 4,096 position-control combinations, including both values of the ignored cell bit. Those checks supply candidate sets before expansion. The original builder and its consumers also execute together for 28 seeded payload profiles, supplied with a rate index and received record. Peer rate agreement, training-bitmap binding and transmitted parameter framing remain open.

8 Downstream mapping organization

8 1 Six positions and two candidate sets

Six two-bit controls select the candidate sets for the six positions. A size-distribution routine assigns one of two set sizes to each position: control zero chooses the high-byte size and controls 1, 2 or 3 choose the low-byte size. The count and size routines match the original instructions for all 4,096 combinations.

Table 10 — Position control behavior

Control	Candidate set	Additional expansion flag
0	First candidate	Clear
1	Second candidate	Set
2	Second candidate	Clear
3	Second candidate	Set

The expansion flag sets the stored sign bit of a working entry. Payload sign selection toggles this bit, so the flag determines the reference polarity for that candidate bank. It does not carry an additional independent payload bit.

8 2 Expansion into working banks

The expansion component constructs six working banks from the two candidate tables. It consumes one control per bank, chooses the corresponding source set and copies its configured length. An odd control sets the additional flag in each copied entry. Successive banks begin 128 words apart; that spacing is a storage stride, not a requirement to use 128 levels.

The expansion uses a last-index convention: a configured value of three copies four entries. The startup caller supplies the value nine, so expansion copies ten entries, whereas the constructor and level-conversion loop each prepare nine. Tests seed the additional entry explicitly. The payload amplitude selector treats the configured size as a radix and selects indices zero through size minus one. It therefore uses only nine entries for radix nine; the copied tenth entry is outside this payload selection range. Its use elsewhere remains open.

8 3 Constructor and consumer execution

The original startup caller has now been executed through candidate construction, level conversion and six-bank expansion in 128 seeded contexts. It sets all six size slots to nine, preserves the selected constructor codewords in the low octets, and copies the transformed candidate entries into the banks selected by the controls. These checks cover the unmodified caller and its callees together.

The second candidate table and additional entries are seeded in those contexts. The result verifies their consumption and the constructor relationship; it does not establish how both candidates are populated in an actual negotiated connection. The intermediate level converter has been executed, but its amplitude and companding interpretation has not been independently reconstructed.

8 4 Verified source placement

A fresh stock controller run copied all seven overlay sources into each of the four modem RAM banks. All 28 byte comparisons matched the corrected source offsets over the complete overlay lengths. The earlier one-word discrepancy was a flat-file extraction error. The downloader does not discard a leading word. This removes the placement qualification from the component results in clauses 7 to 9.

9 Codeword sets and signal realization

9 1 Paired constructor tables

The constructor selects one of the two nine-entry tables in Table 11 according to a local mode flag. Every value in the second table is sixteen lower than the corresponding value in the first. These are alternative constructor tables; they must not be equated automatically with the two candidates consumed by bank expansion. The corrected source placement and original startup caller both support the table contents.

Table 11 — Paired constructor values

Entry	First set	Second set
1	A5	95
2	A7	97
3	AD	9D
4	AF	9F
5	B7	A7
6	BD	AD
7	C5	B5
8	CF	BF
9	E5	D5

All entries are hexadecimal codeword values in stored table order. The difference of sixteen is a relationship between codeword indices. It shall not be converted into a fixed amplitude ratio or assigned to a companding law until the selected law and subsequent conversion path are established.

9 2 From constellation choice to a transmitted sample

The recovered preparation chain selects parameter tables, derives candidate sizes, distributes them across six positions and expands working banks. The preceding mapping block also processes six values and uses auxiliary level tables. Together these establish a concrete six-position preparation architecture, beyond an isolated list of codewords.

Clauses 10 and 11 specify the recovered payload core: source scrambling and bit grouping, mixed-radix amplitude selection, differential signs, minimum-disparity shaping and output formatting. Tests execute the original frame dispatcher and sample steps repeatedly. This establishes the local six-symbol boundary; the signal that establishes that boundary between peers remains open.

9 3 Remaining connection requirements

A complete connection still requires the received training bitmap to be connected to valid rate and constellation choices, the peer alignment procedure to be traced, and the client equalizer and payload decoder to be reconstructed. The larger Quad PCM overlay supplies the payload evidence; its complete x2 selection edge has not been traced. These results shall not be assumed to describe every x2 or V.90 operating branch.

10 Payload framing and amplitude selection

10 1 Parameters and bit organization

A data frame produces six codewords, one for each selected position bank. B specifies the number of amplitude-selection bits. MD specifies the number of independent sign bits, from zero to six. The frame consumes MD sign bits first, followed by B amplitude bits, for a total of B + MD information bits. Within each group, the first consumed bit is the least significant bit of the corresponding integer.

Table 12 — Payload parameters

Symbol	Meaning	Recovered range
B	Amplitude bits consumed per six-symbol frame	19 to 37 in the rate-index table
MD	Independent sign bits per frame	0 to 6
m_0 to m_5	Number of selectable entries in each position bank	Configured separately for each position
X	Unsigned integer represented by the B amplitude bits	Zero through 2^B minus one

The remaining six minus MD signs are shaping choices. They consume no independent source bits. MD equal to six removes this sign-search freedom; MD equal to zero leaves all six signs available for shaping. The tested bank configurations cover rate indices 1 through 14. The final rate-table entry exists, but a valid uniquely decodable configuration for that entry has not yet been established.

10 2 Scrambling and source buffering

Source words are scrambled before entering a circular 128-bit buffer. The recovered source routine supports word widths from one to eight bits. Continuous byte tests use eight-bit words. The modem role flag selects one of two feedback recurrences; the scrambler history continues across byte and frame boundaries. Each scrambled word enters the buffer least significant bit first.

The frame reader also advances least significant bit first and wraps at the end of the buffer. The dispatcher requests more source input unless the available bit count is strictly greater than B + MD. Equality therefore triggers a refill even though the next frame itself would fit. The dispatcher checks availability for every output sample. The stream harness prefills enough bits to retain this margin after consuming a frame.

10 3 Mixed radix amplitude selection

Let X_0 be X. For position i, from zero through five, divide X_i by the configured bank size m_i . The remainder selects the entry in that position bank; the quotient becomes X_{i+1} for the next position. Position zero therefore carries the least significant mixed-radix digit. Bank sizes may differ.

Equivalently, $X = a_0 + m_0 a_1 + m_0 m_1 a_2 + m_0 m_1 m_2 a_3 + m_0 m_1 m_2 m_3 a_4 + m_0 m_1 m_2 m_3 m_4 a_5$, where a_i is the selected entry index and $0 \leq a_i < m_i$. A valid allocation requires the product of the six bank sizes to be at least 2^B . Combinations representing $X \geq 2^B$ are unused amplitude combinations for this allocation.

Each selected entry supplies a reference codeword and an unsigned level index. The level index selects the magnitude used by sign shaping and linear sample monitoring. The codeword carries its own reference polarity, which the sign stage may toggle. Unique decoding requires distinct low-seven-bit reference codewords within each selected bank.

11 Differential signs and disparity shaping

11 1 Independent sign encoding

The MD input sign bits update a continuous binary parity state. Each new parity value is the previous parity exclusive-OR the next input bit. The resulting parity values determine the independent sign toggles. This state persists between frames; resetting it at every frame would change the payload representation.

When MD is less than six, positions are ranked by selected level index, with the largest index first. Equal indices are ordered by position, with the first emitted position taking precedence. The MD highest-ranked positions receive the independent sign toggles. The toggles are assigned to those positions in emission order, rather than in rank order. With MD equal to six, all positions receive them directly in emission order.

11 2 Selection of the remaining signs

The firmware combines the fixed independent signs with a running disparity term, then considers every sign assignment for the remaining positions. It selects the assignment with the smallest absolute resulting disparity. At most 64 assignments are needed when all six signs are free. The level contributions in this search use the recovered sign-toggle convention; the final stored polarity also includes the reference sign already present in the bank entry.

The disparity term equals eight times a signed delayed monitor value. The monitor follows emitted signed linear levels, with each level shifted right by three bits using arithmetic rounding. Its delayed state feeds the following frame's sign decision. Reconstructing only a single isolated frame would omit this dependence.

Free signs are numbered in emission order. The search examines their binary masks from the largest to zero and replaces its choice only for a strictly smaller absolute disparity. Ties therefore select the largest mask. This deterministic tie rule is part of the recovered mapping.

11 3 Codeword and sample output

Each selected sign toggle changes bit seven of its reference codeword. The six resulting octets are emitted in bank order. A configured final exclusive-OR mask, either zero or hexadecimal 2A in the tested profiles, converts the stored representation to the selected output format. It applies after sign selection.

The sample step also converts the selected level to signed linear form, updates the disparity monitor and energy accumulator, applies the negotiated scale, and writes a circular delay buffer. The recovered core and C reconstruction agree on these effects for the ordinary output branch. An optional alternate companding branch selected by another local override remains outside this verified scope.

11 4 Ideal decoding and continuity

Given the six correctly received octets, a decoder first reverses the final format mask. For each bank it matches the low seven codeword bits to a unique reference entry, recovering the amplitude digit and the sign toggle relative to that entry. Combining the digits reconstructs X. The same level ranking identifies which signs carry information; consecutive parity values recover the MD independent sign bits.

The decoder reassembles the sign bits followed by the amplitude bits, carries partial bytes across frame boundaries, and reverses the role-selected scrambler. This ideal symbol-domain inverse recovers the original source bytes in the continuous tests. It assumes that the six octets, bank configuration, frame alignment and initial parity are already known. It does not implement an analogue equalizer, acquisition procedure or the recovered client receiver.

12 Symmetric role and capability selection

12 1 Roles and compatible endpoints

Vendor management definitions identify x2 symmetric as a distinct role: both endpoints send and receive at the same x2 speed. The HiPer DSP pairing diagnostic permits either a server with a client, or two symmetric endpoints. An asymmetric server setting alone therefore does not establish compatibility between two digital modems.

Symmetric operation requires digitally attached endpoints and a channel that preserves the required sample alphabet in both directions. A conventional analogue Courier has a local codec and DAA; its shared 64000 result strings do not establish symmetric support. The I-modem exposes server and symmetric controls in S58. Total Control equipment exposes corresponding disable controls in S76. These register numbers and values are product specific.

Table 13 — I-modem S58 controls

Value	Meaning when set	Symmetric significance
1	Disable x2	Reject x2 operation
2	Disable asymmetric server mode	Does not itself disable symmetric mode
4	Force x2 A-law mode	Select the advertised law option
8	Disable symmetric mode	Remove symmetric capability
16	Enable -6 dBm constellation	Optional advertised constellation
32	Disable V.90	Keep the tested call on x2

Values 1, 2 and 8 are disable controls, as confirmed by the decoded help tokens and capability builder. For the recorded I-modem test, S58 = 48 sets values 16 and 32, leaving x2, server and symmetric operation enabled. S58 = 34 leaves symmetric operation enabled while disabling asymmetric server mode and V.90; the earlier restricted-DTE tests of that setting reached V.34, not symmetric x2.

12 2 Capability advertisement and mode selection

Setting the server-disable control clears capability-word bit 12; setting the symmetric-disable control clears bit 11. The constellation option sets bit 10. The capability word is transferred to the transmit INFO0 work area. This establishes the control-to-advertisement path, but does not recover the complete peer decision that selects symmetric operation.

Capability advertisement, enabled-rate mask and datapump mode are separate parameters. The actual engine mode comes from call descriptors, through a different command path. A capability bit does not by itself identify the active engine branch. The fixed digital-call controls in S67 also operate separately from S58 symmetric capability. Annex A records the corrected command identities and source references.

13 Symmetric bearer and sample transforms

13 1 Digital rate and word width

At an 8000-word/s bearer cadence, eight information bits per word give 64000 bit/s in each direction; seven give 56000 bit/s. The recovered I-modem digital width routine installs matching transmit and receive widths and masks. Mode bit zero clear selects eight bits and mask FF hexadecimal; set selects seven bits and mask 7F hexadecimal. The fixed digital-call rate control in S67 reaches this mode through the call-descriptor command path.

Digital 56000 shall be distinguished from analogue downstream x2 56000. The former is seven bits per digital bearer word. The latter is a six-symbol allocation of 42 information bits over 8000 samples/s, using mapped codewords and the sign machinery in clauses 10 and 11. Equal printed rates do not imply identical coding.

The width and sample routines are verified components of the server and digital-mode program. A live Ie030002 pair trace confirms the eight-bit digital callbacks, GPC scrambling and octet reversal during the successful symmetric 64000 call. The six-symbol Quad downstream mapper shall therefore not be assigned to symmetric 64000 simply by extrapolating its rate table.

13 2 Transmit and receive processing

The recovered transmit callback obtains a source word, masks it to the selected width, optionally scrambles it, optionally reverses the octet and writes the sample. The receive callback records the incoming sample, reverses the octet when selected, masks it and optionally descrambles it before calling the sink. Reversal changes bit order within a word; it does not change the number of information bits.

For the directly called scrambler and descrambler entries, the feedback recurrence is $1 + x^{-18} + x^{-23}$, known as GPC. If $u[n]$ is the input bit and $s[n]$ the transmitted scrambled bit, $s[n] = u[n] \text{ XOR } s[n-18] \text{ XOR } s[n-23]$. Descrambling uses the received scrambled history in the same XOR relation to recover $u[n]$. Execution checks cover the updated history as well as the resulting words. The successful Ie030002 64000 pair also verifies activation of both transforms and zero-history initialization before the data phase.

Transmit and receive histories are separate and continue across callback invocations. The component tests preload a source word and disable source and sink dispatch. The separate live-call trace samples original callbacks with source and sink dispatch running; complete queue semantics and symmetric acquisition remain open. An additional width-aware training source packs repeated 7E hexadecimal into a bit reservoir; 256 consecutive words at each of the seven-bit and eight-bit widths match the original instructions.

13 3 Channel preservation

The successful pair exchanges PCMU bearer octets directly between two native I-modem datapumps. Its result applies to that byte-exact digital path. Linear-audio conversion, companding-law changes, resampling, lossy transport or robbed-bit signalling can alter the available alphabet and require separate qualification. An 8000-sample/s audio path alone does not establish eight-bit transparency.

14 Symmetric connection evidence and limits

14 1 Recorded full-rate pair

Two original Ie030002 supervisors and native C51 datapumps establish CONNECT 64000/ARQ/x2/LAPM over a direct PCMU B-channel. Both use S54 = 0, S58 = 48, the saved National ISDN-1 profile, a 230400-bit/s DTE and an unrestricted PCM rate mask.

Table 14 — Recorded pair outcomes

Configuration	Carrier at both ends	Bidirectional data
S54=0S58=32	64000/ARQ/x2/LAPM	No markers delivered; carrier lost
S54=0S58=48	64000/ARQ/x2/LAPM	Both distinct markers delivered exactly

The successful run records 252 complete 100 ms bearer frames per endpoint, zero PCM underruns and no CPU or DSP error. The caller receives ANSWER-X2-S58-48 and the answerer receives CALLER-X2-S58-48. Original firmware produces carrier and DTE output; the harness injects neither.

This emulator test verifies carrier and short bidirectional payload delivery. Sustained throughput, hardware interoperability and public-network operation remain untested. The S58 = 32 control reaches carrier but loses it without delivering markers; the test does not establish S58 = 48 as a universal peer requirement.

14 2 DTE and rate eligibility

The originating supervisor rejects x2 below the 38400-bit/s DTE entry. It restricts the PCM rate mask to 0003 through the 57600 entry and to 0FFF at 115200; 230400 keeps the full mask. These are firmware-profile limits. Early 9600-bit/s DTE tests reached V.34.

14 3 Live full rate scrambling evidence

A repeat pair trace uses the original Ie030002 transmit callback E97E and receive callback E992. The caller engine mode is 0310 hexadecimal and the answerer mode is 0210; both set scrambling mask 0200. Both transmit widths are eight, with mask FF, and FFD9 bit 15 enables octet reversal. The verified digital chain therefore assembles eight source bits, applies GPC scrambling, reverses the octet and writes one bearer word. These source bits include the modem link framing; raw terminal octets are not sent unchanged onto the bearer.

The live trace captures zero transmit and receive histories at E908 twice per endpoint: during initial mode setup and again after the data callbacks are selected, before CONNECT. The width initializer E8E1 clears transmit cells 03D8 and 03D9 and receive cells 031E and 031F. After CONNECT, 12224 transmit input snapshots provide 12033 consecutive within-window GPC history transitions, all matching the recovered recurrence. The windows are samples of the running call, not a complete bearer recording. Both distinct terminal markers arrive and both receivers report zero underruns.

14 4 Remaining protocol reconstruction

Peer role selection from INFO0, startup ordering, rate acknowledgement, alignment, complete queue semantics and retrain or fallback behavior remain open. The component lift uses IM020104; the live trace uses Ie030002. Their instructions have not been shown identical.

Total Control documentation names V.90 all-digital as DDPCM with separate controls and reported modulation. That metadata does not establish an I-modem implementation. This draft specifies the supported x2 evidence.

15 Rates and information carried per sample

15 1 Rate families

A modem rate describes information bits carried per second, not the size of the stored sample alone. Every six-symbol PCM frame still emits six octets at an 8000-sample/s cadence. Lower information rates use fewer selectable combinations of those octets. In the recovered digital callback, by contrast, the selected width directly specifies seven or eight information bits per bearer word. Both paths can report 56000, but their mappings differ.

Table 15 — Complete x2 reported rate ladder

Reported bit/s	K per six samples	Rate interpretation
33333	25	Six-symbol PCM allocation
37333	28	Six-symbol PCM allocation
41333	31	Six-symbol PCM allocation
42666	32	Six-symbol PCM allocation
44000	33	Six-symbol PCM allocation
45333	34	Six-symbol PCM allocation
46666	35	Six-symbol PCM allocation
48000	36	Six-symbol PCM allocation
49333	37	Six-symbol PCM allocation
50666	38	Six-symbol PCM allocation
52000	39	Six-symbol PCM allocation
53333	40	Six-symbol PCM allocation
54666	41	Six-symbol PCM allocation
56000	42	Six-symbol PCM allocation
57333	43	Six-symbol PCM allocation
64000	48	Digital endpoint; eight bits per word

For the six-symbol interpretation, $R = K \times 8000 \div 6$ bit/s. Printed rates truncate fractional values: 33333 denotes $33333\frac{1}{3}$ and 42666 denotes $42666\frac{2}{3}$. The x2 ladder skips $K = 26, 27, 29$ and 30 ; it is not every integer allocation between its lowest and highest entries. Rates below 33333 belong to another modulation or digital rate-adaptation mode in the inspected product, not to this x2 ladder.

Table 15 records the firmware result ladder and its numerical six-sample correspondence. It does not establish a symmetric call at every listed rate, nor a wire encoding for those rates. $K = 48$ at 64000 is an arithmetic comparison, not evidence that the lower-rate constellation mapper produces that digital endpoint.

16 Lower rate PCM mapping and scrambling

16 1 Information allocation

The recovered Quad PCM frame consumes $K = B + MD$ scrambled information bits: MD bits for independent signs, followed by B bits for amplitude choices. MD can be zero through six. The six minus MD remaining signs carry shaping choices rather than additional source bits. The nominal information rate for a given allocation is $(B + MD) \times 8000 \div 6$. The sample cadence remains 8000/s.

A reduction in B restricts the amplitude-choice integer and the required bank capacity. A reduction in MD leaves more signs free for disparity shaping. Neither operation amounts to taking the low five or six bits of each input byte and sending them directly. Byte boundaries can cross frames; scrambler history and differential-sign parity continue across those boundaries.

Table 16 — Executed Quad builder profiles

Local index	B	Six bank sizes in emission order	Rate if MD is 6
1	19	9, 9, 9, 9, 9, 9	33333
2	22	12, 13, 13, 13, 13, 13	37333
3	25	18, 18, 18, 18, 18, 18	41333
4	26	21, 20, 20, 20, 20, 20	42666
5	27	27, 22, 22, 22, 22, 22	44000
6	28	28, 25, 25, 25, 25, 25	45333
7	29	28, 29, 29, 29, 29, 29	46666
8	30	32, 32, 32, 32, 32, 32	48000
9	31	41, 35, 35, 35, 35, 35	49333
10	32	42, 40, 40, 40, 40, 40	50666
11	33	47, 45, 45, 45, 45, 45	52000
12	34	50, 51, 51, 51, 51, 51	53333
13	35	52, 62, 62, 62, 62, 62	54666
14	36	64, 64, 64, 64, 64, 64	56000
15	37	Valid profile remains open	57333

Indices 1 through 14 are executed, seeded builder contexts, with matching profiles for both tested format flags. Index 14 uses mode two; the others use mode zero. Index 15 has a stored B value of 37, but a uniquely decodable constructed profile has not been verified. The right-hand column is calculated with MD = 6; it is not a captured rate agreement. A local builder index shall not be used directly as a transmitted rate code.

16 2 Worked allocations

Local index 8 supplies B = 30 and six banks of 32 entries. With MD = 6, the frame carries 36 information bits and corresponds to 48000 bit/s. With MD = 3, it carries 33 bits and corresponds arithmetically to 44000 bit/s, with three signs reserved for shaping. With MD = 0 it corresponds to 40000 bit/s, which is absent from the x2 result ladder. These examples show why component arithmetic alone does not establish a negotiable rate.

17 Data transforms and lower rate symmetric limits

17 1 Lower rate transmit and receive chain

For the verified Quad PCM core, the transmit order is source bits, role-selected scrambling, bit buffering, sign and amplitude grouping, mixed-radix amplitude selection, differential signs and disparity shaping, then output formatting. Scrambling occurs before mapping. The source routine calls its scrambler as part of every append in the verified branch; it is not an optional post-mapping octet XOR.

The role flag selects the recovered GPC recurrence $1 + x^{-18} + x^{-23}$ or GPA recurrence $1 + x^{-5} + x^{-23}$. Continuous tests cover both role values. The feedback history is the transmitted scrambled bit stream; the receiver uses received scrambled history for the inverse. The modem-role assignment of every complete call branch and its scrambler reset timing remain separate reconstruction work.

After undoing output formatting, an ideal receiver recovers bank digits and reference-relative signs, reverses mixed-radix selection, recovers the independent bits from differential parity, reassembles the sign and amplitude groups into a continuous bit stream, then descrambles that stream into source bytes. Analogue acquisition and equalization are needed before these symbol decisions. The ideal inverse used in the tests assumes exact octets and known alignment.

17 2 Direct digital transforms

The I-modem digital callback masks each assembled word, conditionally scrambles it with GPC, conditionally reverses its octet and writes it. Engine mode mask 0200 hexadecimal enables scrambling. A separate flag selects reversal. Reception reverses the order of those transforms: optional reversal, width mask and optional GPC descrambling. Eight-bit mode gives 64000 bit/s; seven-bit mode packs a continuous source stream into seven-bit words and gives 56000 bit/s. It does not drop every eighth source bit.

No five-bit, six-bit or fractional-width selection has been recovered in that width initializer. Lower x2 rates therefore shall not be described as direct words of progressively fewer bits based on this routine. Likewise, optional scrambling in this callback shall not be confused with the mandatory source scrambler in the tested Quad mapping branch. The successful 64000 symmetric pair has now been traced: eight-bit digital callbacks, GPC scrambling and octet reversal are active on both ends. Seven-bit symmetric activation and lower-rate call profiles still require their own live traces.

17 3 What lower rate symmetric operation establishes

Vendor symmetric mode requires the same x2 rate in both directions. The saved successful test establishes 64000/64000. No saved full-call test used here establishes 56000/56000 or a lower symmetric rate, and the recovered lower-rate Quad mapper has not been tied to a complete symmetric call. Digital seven-bit operation is established as a component and control path, not as proof of symmetric x2 at 56000.

For a complete lower-rate symmetric specification, a call must connect the negotiated directional rates to the selected transmit and receive callbacks, word width or six-symbol allocation, bank records, scrambler flags and initial states. Bidirectional payload must then be recovered at that rate. V.34 fallback and the I-modem V.110, V.120 or X.75 rate-adaptation paths use different coding and shall not be counted as lower-rate x2 merely because they share a B-channel or a result speed.

Annex A Firmware evidence and reproduction

This annex is informative. It retains implementation detail for traceability without making addresses part of the protocol description. The manually reconstructed C is not vendor source.

Table A 1 — Firmware profiles

Profile	Image	Role
Client	3453C version 2.3.33 XMF	Retained x2 setup, low overlay and PCM training
Original server	IM020104.NAC	Capability packing, startup source and sample transforms
Later server	QF060003.NAC	PCM preparation and six-symbol payload core

Evidence directories under artifacts are 3453c-x2-decomp-20261004, x2-server-decomp-20261004, x2-mapper-negotiation-20261004 and x2-payload-mapper-20261004. Each preserves the reconstruction and comparison evidence for its profile.

Table A 2 — Comparison coverage

Group	Count	Qualification
Client	1,769	Original mapped client instructions
Original server	6,474	Original mapped server instructions
Quad helpers	88	Corrected stock-verified source placement
Mapper C comparisons	13,504	Transfer, unpacking, count, sizes and bank expansion
Startup caller contexts	128	Unmodified constructor and consumers; seeded candidates
Overlay source copies	28	Seven overlays across four stock controller RAM banks
Payload core	9,924	Components, 28 builder profiles and continuous output
Decoded source bytes	3,920	28 streams of 32 frames with an ideal symbol inverse

Mapper entries are AB06 for received-record transfer, C870 for unpacking, C6E5 for the six-position count, C6F4 for size distribution and C92A for descriptor expansion. The first candidate source is 0A60 and the second is 0880; expanded banks begin at E8F4 with a stride of 0080. The codeword constructor is C9BF; the integrated startup caller begins at C840.

Run tools/recover_3453c_x2.py, tools/recover_x2_server.py, tools/recover_x2_mapper.py and tools/recover_x2_payload.py with the repository Python environment. The scripts compile manual C lifts and compare their effects with original DSP instructions. Run tools/verify_quad_pcm_placement.py for the independent stock overlay-copy check.

The payload entries are B2BA for source append, C5DF for amplitude selection, C642 for signs and C544 for frame output. Builder entry C862 receives a preselected rate index. The tested branch has 039F bit 7 and FFD9 bit 12 clear. Profiles use default position choices, flags zero and one, mode zero for rate indices 1 through 13 and mode two for index 14. verification.json and example.json preserve the results. The payload aggregate counts component cases, sample steps and streams; it does not represent independent connection trials.

The protocol history in docs/x2-v90-protocol-selection.md includes the preamble-detector correction. Field terminology follows ITU-T V.34 Table 14, available at <https://www.itu.int/rec/T-REC-V.34-199610-S/en>. V.90 context is at <https://handle.itu.int/11.1002/1000/4493>.

Draft 0.7 confirms active GPC scrambling, octet reversal, eight-bit digital callbacks and pre-data zero-history resets in the successful symmetric 64000 pair. Draft 0.6 added the complete x2 rate ladder, per-frame information accounting and executed lower-rate bank profiles. Draft 0.5 added symmetric controls and the recorded 64000/64000 pair. Complete peer negotiation, the final analogue rate-table profile, alternate companding output and analogue client decoding remain open. Forced higher-rate records can produce duplicate low-seven-bit entries or fail to terminate in the builder; they do not establish valid negotiated profiles.

Symmetric role and control evidence is collected in docs/x2-symmetric-and-all-digital.md, especially its later command-table, bit-number and tokenized-help corrections. Vendor sources are HDM.MIB in docs/x2/tcmw551.zip, the HiPer DSP reference 24187300.PDF and the NETServer RADIUS dictionary in docs/x2/ne050303.zip. The management controls are S76.0 client-disable, S76.1 server-disable and S76.2 symmetric-disable; the dictionary reports x2symmetric as modulation 30. These identifiers are management metadata, not wire fields.

In IM020104, host tags 70 and 71 reach DSP entries 91A1 and 91AE, storing capability and rate words at FFDB and FFDA. Mode tags 5C and 5D reach E11E and E11A. Digital width setup is E5E7; sample callbacks are E65F and E67B; the directly called GPC entries are 8FAA and 8FE1; octet reversal is E7E7. The corrected dispatch-table base is 882B. Tag 70 is not the engine-mode command.

The full-call evidence is artifacts/imodem-pair-x2-full-rate-high-power-20261002/verification.json; the carrier-only control is artifacts/imodem-pair-x2-full-rate-v8-server-20261002/verification.json. Both successful endpoints send 0070:3DFF and 0071:FFFF. Setup, DTE restrictions and the reproduction command are documented in docs/imodem-x2-pair.md. The paired harness is tools/probe_imodem_pair.py; its --check-x2 gate requires x2 in the firmware CONNECT result. Marker checks independently establish data delivery.

The live scrambler evidence is artifacts/x2-scrambler-call-reset-20261004/verification.json and the associated originate.json and answer.json traces. tools/probe_imodem_pair.py --trace-digital-path records mode writes and read-only instruction-boundary snapshots; tools/verify_x2_scrambler_trace.py checks widths, flags, callback addresses, zero histories and the GPC recurrence. These addresses apply to Ie030002, not to the earlier IM020104 lift. Initial connection resets are verified; retrain and lower-rate resets remain open.

The Quad B table is program C9E0 through C9EE: 19, 22, 25, then 26 through 37. C862 subtracts one from the local index before looking up this table. MD is read separately from bits 8 through 10 of the working position record. Table 16 is generated from the saved payload verification profiles; only its final unverified row uses the stored table value directly. Continuous stream checks use B = 32 with all MD values, both role flags and both tested output-format states.